

P1.9 Energy and power

Learning objectives

After this topic, you should know:

- what is meant by power
- how to calculate the power of an appliance
- how to calculate the efficiency of an appliance in terms of power
- how to calculate the power wasted by an appliance.

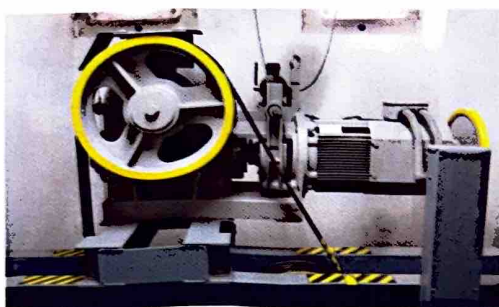


Figure 1 A lift motor

Worked example

A motor transfers 10 000 J of energy in 25 s. Work out its power.

Solution

$$P = \frac{E}{t}$$

$$P = \frac{10\,000\text{ J}}{25\text{ s}} = 400\text{ W}$$

Study tip

Always use the correct units in your calculations. Values that have been converted to micro (μ), milli (m), kilo (k), or mega (M) must be converted back to the SI unit. Standard form is useful when working with large numbers (See Maths skills 1b for more about standard form.).

When you use a lift to go up, a powerful electric motor pulls you and the lift upwards. The electric current through the lift motor transfers energy to the gravitational potential energy store of the lift when the lift goes up at a steady speed. You also get energy (from the electric current) transferred to the thermal energy store of the motor and the surroundings due to friction between the moving parts of the motor. In addition, energy is transferred to the thermal energy store of the surroundings by sound waves created by the lift machinery.

- The energy you supply to the motor per second is the **power** supplied to it.
- The more powerful the lift motor is, the faster it moves a particular load.

The more powerful an appliance is, the faster the rate at which it transfers energy.

The power of an appliance is measured in watts (W) or kilowatts (kW).

1 watt is equal to the rate of transferring 1 joule of energy in 1 second (i.e., $1\text{ W} = 1\text{ J/s}$)

1 kilowatt is equal to 1000 watts (i.e., 1000 joules per second or 1 kJ/s).

You can calculate power using the equation:

$$\text{power, } P \text{ (watts, W)} = \frac{\text{energy transferred to appliance, } E \text{ (joules, J)}}{\text{time take for energy to be transferred, } t \text{ (seconds, s)}}$$

Power ratings

Table 1 shows some typical values of power ratings for different energy transfers.

Table 1

Appliance	Power rating
A torch	1 W
An electric light bulb	100 W
An electric cooker	10 000 W = 10 kW (where 1 kW = 1000 watts)
A railway engine	1 000 000 W = 1 megawatt (MW) = 1 million watts
A Saturn V space rocket	100 MW
A very large power station	10 000 MW
World demand for power	10 000 000 MW
A star like the Sun	100 000 000 000 000 000 000 MW

Orders of magnitude

The power ratings in Table 1 are called 'order of magnitude' values. This means that the values are estimates to the nearest power of ten.

The symbol $\sim$ is used for an order of magnitude estimate. For example, if ten million people switch their electric kettles on at the same time (e.g., at half-time in a big TV football match), the jump in the demand for electric power would be $\sim 10\,000\text{ MW}$ (10^4 MW).

Muscle power

How powerful is a weightlifter?

A 30 kg dumbbell has a weight of 300 N. Raising it by 1 m would increase its gravitational potential energy store by 300 J. A weightlifter could lift it in about 0.5 seconds. The rate of energy transfer would be 600 J/s ($= 300 \text{ J} \div 0.5 \text{ s}$). So the weightlifter's power output would be about 600 W in total! The power output of two weightlifters could be compared by measuring how long they each take to lift the same weight through the same vertical height. The energy transferred is the same. So the one who takes the less time has a bigger power output.

Efficiency and power

For any appliance:

- its useful power out (or output power) is the useful energy per second transferred by it
- its total power in (or input power) is the energy per second supplied to it.

In Topic P1.7, you learnt that, for an appliance:

$$\text{efficiency} = \frac{\text{useful energy transferred by the device}}{\text{total energy supplied to it}} (\times 100\%)$$

Because power = energy per second transferred or supplied, this efficiency equation can be rewritten as

$$\text{efficiency} = \frac{\text{useful power out}}{\text{total power in}} (\times 100\%)$$

Wasted power

In any energy transfer, the energy wasted = the input energy supplied – the useful output energy. Because power is energy transferred per second:

$$\text{power wasted} = \text{total power in} - \text{useful power out}$$

- State which of the following is more powerful. [1 mark]
 - A torch bulb or a mains filament lamp. [1 mark]
 - A 3 kW electric kettle or a 10 000 W electric cooker. [1 mark]
 - There are about 20 million occupied homes in England. If a 3 kW electric kettle was switched on in 1 in 10 homes at the same time, work out how much power would need to be supplied. [1 mark]
- The total power supplied to a lift motor is 5000 W. In a test, the motor transfers 12 000 J of energy in 20 s to the gravitational potential energy store of the lift.
 - Calculate how much energy is supplied to the motor by the current through it in 20 s. [2 marks]
 - Calculate its efficiency in the test. [2 marks]
- A machine has an input power rating of 100 kW. If the useful energy transferred by the machine in 50 seconds is 1500 kJ, calculate:
 - its output power in kilowatts [2 marks]
 - its percentage efficiency. [2 marks]
- Describe the energy transfers and changes to the energy stores of an electric hot-water shower when it is in operation. [5 marks]
 - A 12 kW electric shower is used 4 times in a day for 20 minutes each time. Calculate the energy supplied to it by electricity in one day. [2 marks]

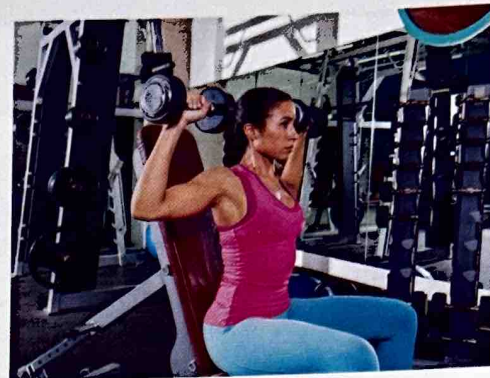


Figure 2 Muscle power

Worked example

The useful power out of an electric motor is 20 W, and the total power into it is 80 W. Calculate:

- the percentage efficiency of the motor
- the power wasted by the motor.

Solution

- percentage efficiency = $\frac{\text{useful power out}}{\text{total power in}} \times 100\%$
 $= \frac{20 \text{ W}}{80 \text{ W}} \times 100\% = 25\%$
- wasted power = $80 \text{ W} - 20 \text{ W} = 60 \text{ W}$

Key points

- Power is rate of transfer of energy.
- The power of an appliance is $P = \frac{E}{t}$.
- efficiency of an appliance
 $= \frac{\text{useful power out}}{\text{total power in}} (\times 100\%)$
- power wasted by an appliance = total power input – useful power output